Mutual simulation of Hamiltonian dynamics on n interacting quantum systems

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Control-theoretical model of quantum computers

Time evolution defined by the time-dependent Schrödinger equation

$$\frac{d}{dt}U(t) = -iH(t)U(t)\,, \quad U(0) = \mathbf{1}$$

$H(t)$ Hamiltonian

$U(t)$ is the resulting unitary after time t

$$H(t) = H_d + \sum_{i=1}^{m} a_i(t)H_i$$

H_d drift Hamiltonian internal to the system

$\sum_{i=1}^{m} a_i(t)H_i$ control Hamiltonian externally changeable

Complexity of a unitary U: minimal time required to implement U

Fast control limit

strength of the control Hamiltonians arbitrarily large $\Longrightarrow$

control unitaries arbitrarily fast

Complexity of a unitary U: minimal time

$$U = \exp(-iHt_N)V_N \cdots \exp(-iHt_2)V_2 \exp(-iHt_1)V_1$$

express it as the evolution according to the piecewise constant Hamiltonian

$$U = U_N \exp(-iU_N^\dagger H U_N t_N) \cdots \exp(-iU_2^\dagger H U_2 t_2) \exp(-iU_1^\dagger H U_1 t_1)$$

Average Hamiltonian Theory

Dyson expansion

$$U(t) = \mathcal{T}\exp(\int_{\tau=0}^{t} d\tau H(\tau)) = \exp(-i\bar{H}t)$$

Magnus expansion $\bar{H} = \bar{H}^{(0)} + \bar{H}^{(1)} + \cdots$

$$\begin{aligned} \bar{H}^{(0)} &= \frac{1}{t}\int_0^t d\tau H(\tau)\,, & \|\bar{H}^{(0)}\| \leq 1 \\ \bar{H}^{(1)} &= \frac{-i}{2t}\int_0^t d\tau' \int_0^t \tau''[H(\tau'), H(\tau'')]\,, & \|\bar{H}^{(1)}\| \leq t/2 \end{aligned}$$

$\Longrightarrow U(t)$ essentially determined by $\bar{H}^{(0)}$ for small t

Simulating Hamiltonians/First order simulation

H can simulate $\tilde{H}$ with overhead 1, written $\tilde{H} \prec H$, iff

$$\tilde{H} = \sum_{j=1}^{N} p_j U_j^\dagger H U_j$$

with $\sum_j p_j = 1$ and U_j control operations, i.e.

$\tilde{H}$ is a convex sum of unitary conjugates of H

H can simulate $\tilde{H}$ with overhead τ iff $\tilde{H} \prec \tau H$

time-optimal simulation problem $\longrightarrow$ convex optimization problem

Quantum Networks

system consisting of n nodes $\quad \mathcal{H} = \mathbb{C}^d \otimes \mathbb{C}^d \otimes \cdots \otimes \mathbb{C}^d$

pair-interaction Hamiltonian

$$H = \sum_{k<l} \sum_{\alpha\beta} J_{kl;\alpha\beta} \sigma_\alpha^k \sigma_\beta^l \, , \quad \sigma_\alpha \text{ ONB of } su(d)$$

coupling matrix

$$J = \left(\begin{array}{c|c|c|c|c} 0 & J_{12} & J_{13} & \cdots & J_{1n} \\ \hline J_{21} & 0 & J_{23} & \cdots & J_{2n} \\ \hline \vdots & \vdots & & & \\ \hline J_{n1} & J_{n2} & J_{n3} & & 0 \end{array} \right) \in \mathbb{R}^{(d^2-1)\, n \times (d^2-1)\, n}$$

control operations are all local operations in

$$SU(d) \otimes SU(d) \otimes \cdots \otimes SU(d)$$

Decoupling/Switching off the time evolution

Conjugating by unitaries of an error basis $\mathcal{E} = \{U_1 = \mathbf{1}, U_2, \ldots, U_{d^2}\}$

$$\sum_j U_j^\dagger \, a \, U_j = \mathbf{0} \qquad \text{for all } a \in su(d)$$

cancels all matrices in $su(d)$

Combinatorial problem: which sequences of conjugations allow to switch off all interactions?

Efficient solution: decoupling in networks based on orthogonal arrays

number of conjugations grows linearly with n

Action of control operations on the coupling matrix J

Conjugation of the pair-interaction Hamiltonian H by control operations

corresponds to

conjugation of J by a direct sum of orthogonal matrices

$$\begin{aligned} H &\mapsto (u_1 \otimes u_2 \otimes \cdots \otimes u_n)^\dagger H (u_1 \otimes u_2 \otimes \cdots \otimes u_n) \\ J &\mapsto (U_1 \oplus U_2 \oplus \cdots \oplus U_n) J (U_1 \oplus U_2 \oplus \cdots \oplus U_n)^T \end{aligned}$$

Lower bounds on time overhead

If H can simulate $\tilde{H}$ with overhead τ then

$$\tilde{J}/\tau = \sum_{j=1}^{N} p_j (U_{j1} \oplus U_{j2} \oplus \cdots \oplus U_{jn})\, J\, (U_{j1} \oplus U_{j2} \oplus \cdots \oplus U_{jn})^T$$

convex sum of conjugates of J

$\Rightarrow$ necessary condition given by majorization criterion:

the spectrum $\mathrm{Spec}(\tilde{J})$ majorizes the spectrum $\mathrm{Spec}(\tau J)$

Application I: Time reversal

Time reversal: $H \mapsto -H$ (refocusing, spin echoes)

dipole-dipole coupling between all nodes (easy to invert)

$$H = \sum_{k<l} w_{kl}(3\,\sigma_z \otimes \sigma_z - \sum_{\alpha=x,y,z} \sigma_\alpha \otimes \sigma_\alpha)\,, \quad w_{kl} \neq 0$$

lower bound on time overhead is 2

this bound is tight (well known-scheme in NMR)

Application I: Time reversal

strong scalar coupling between all nodes (difficult to invert)

$$H = \sum_{k<l} w_{kl}(\sigma_x^k \sigma_x^l + \sigma_y^k \sigma_y^l + \sigma_z^k \sigma_z^l)\,, \quad w_{kl} \neq 0$$

lower bound on the overhead $n-1$

the inverted time-evolution is necessarily slower by the factor $(n-1)$

upper bound $cn-1$ (decoupling scheme based on orthogonal arrays)

Application II: Hierarchy of Hamiltonians

$$H_1 = \sum_{kl} w_{kl}(\sigma_x^k \otimes \sigma_x^l + \sigma_y^k \otimes \sigma_y^l)$$

$$H_2 = \sum_{kl} w_{kl}(\sigma_x^k \otimes \sigma_x^l - \sigma_y^k \otimes \sigma_y^l)$$

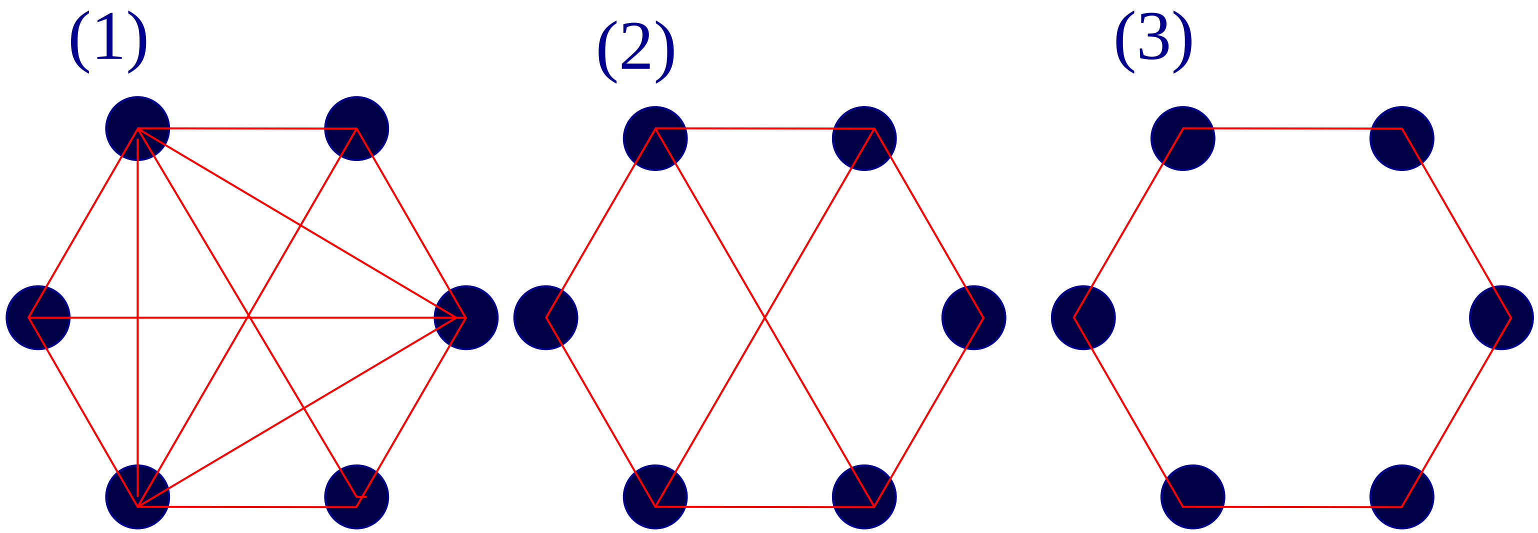
H_1 cannot simulate H_2 with overhead less than $(n-1)$

H_2 can simulate H_1 with overhead 2

Application III: Switching off some interactions

Combinatorial problem: cancel some interactions

Are the other interactions necessarily disturbed?



Yes, for some graphs $\rightarrow$ lower bound on time overhead given by eigenvalues of the adjacency matrix

Upper bound given by chromatic index of interaction graph

Optimal simulation problem equivalent to separability problem

complete zz-Hamiltonian H: interaction between all nodes are of the form $\sigma_z \otimes \sigma_z$

A Hamiltonian with coupling matrix J can be simulated with overhead 1 iff there is a separable n-qubit state ρ such that J is the $3n \times 3n$-matrix

$$\big(tr(\rho\sigma_\alpha^k\sigma_l^l)\big)_{kl\alpha\beta}$$

maximal strength of pair-interactions implementable with overhead 1

corresponds to

maximal strength of pair-correlations in a separable state

Universal simulation using finite control groups

Universal simulation does not require all operations in $SU(d)$

Certain subgroups are sufficient; called transformer groups

Finite transformer groups are characterized in terms of group characters

Conclusions

- Upper and lower bounds on simulation complexity are given by graph-theoretical notions like chromatic index, chromatic number and graph spectra
- Efficient decoupling schemes can be constructed by using orthogonal arrays and error basis
- Universal simulation does not require the ability to perform all local unitary operations; we have found finite control groups that are sufficient

Simulating Arbitrary Pair-interactions by a Given Hamiltonian: Graph-theoretical Bounds on the Time Complexity, quant-ph/0106077, to appear in QIC

Simulating Hamiltonians in Quantum Networks: Efficient schemes and Complexity Bounds, quant-ph/0109088, to appear in Phys. Rev. A

Universal Simulation of Hamiltonians Using a Finite Set of Control Operations, quant-ph/0109063, to appear in QIC

Complexity of inverting n-spin interactions: Arrow of time in quantum control, quant-ph/0106085

Lower Bound on the Chromatic Number by Spectra of Weighted Adjacency Matrices, cs.DM/0112023